

When the Buffer Is the Policy

Six supply chain engagements on rules that were written during a bad week, never reopened, and still running the business.

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EXECUTIVE CONCLUSION

Ask a planning organization for its policies and you get a document. Ask where its decisions come from and you get safety stock settings, planning fences, min/max levels, expedite approvals, and a spreadsheet on somebody’s second monitor.

Those settings are the policy. Most were written in the middle of a bad week by whoever was in the room, and most have not been opened since. A rule built under pressure and left to run in calm conditions behaves very differently from one built in calm conditions to run under pressure. Supply chains are full of the first kind, and the analytics budget almost always goes somewhere else.

01	02	03
Ask when each buffer was set and what was going wrong that week. The answer explains more than the data will.	Every buffer carries a price. Recover it, say it out loud in a room with finance, and watch who disagrees.	Plan stability and expediting are policy choices. Leave them out of the objective and the solver spends them for free.

On these cases

Six engagements are described here, from distribution, discrete manufacturing, retail, consumer goods, and field service. Names are gone, two are welded together from more than one client, and any number specific enough to identify a company has been changed.

Six engagements make a caseload rather than a sample. Selection was never random, and consulting work skews toward companies that already suspect something is off, so treat what follows as a hypothesis with some mileage on it. What gives it weight is that it turned up in shops with different industries, different planning systems, different maturity, and different complaints.

1. Where the policy actually lives

Demand moves. Suppliers slip. Yields wander, trucks miss appointments, and a customer triples an order in the last week of a quarter. Planning moves that variability around the network. Somebody still ends up absorbing it, and there are five candidates for the job.

Inventory. Safety stock, cycle stock, in-transit, consignment, and the pallet a branch manager keeps behind a curtain.

Capacity. Overtime, a second shift, equipment held idle on purpose, contract labor, reserved carrier space.

Time. Quoted lead times, frozen horizons, planning fences, and the days a planner adds to a promise date without telling anyone.

Cash. Expedite premiums, spot freight, air instead of ocean, concessions to keep an account calm.

The customer. Backorders, substitutions, split shipments, partial fills, and dates that move without a conversation.

Every supply chain runs all five. The proportions are rarely chosen. They are sediment: whoever controlled a budget during the last serious disruption decided which absorber got funded, the

number went into a system, and the reasoning walked out with the person.

So the first exercise on most engagements now happens before any modeling. Take one recurring decision with money on it. For each source of variability that touches it, ask where the variability lands, who owns that absorber, and what a unit of it costs.

Four of the five answers come back inside a week. Inventory sits on the balance sheet, overtime is in the labor report, expedite premiums land in freight accruals, and quoted lead times live in the order promising rules. The fifth line takes months, because the cost of a missed date usually lives in one senior person’s memory, and the cost of a substitution the customer swallowed this quarter lives nowhere at all. Build a model on the four you can see and it will push variability onto the one you cannot, which is where the solver will go looking.

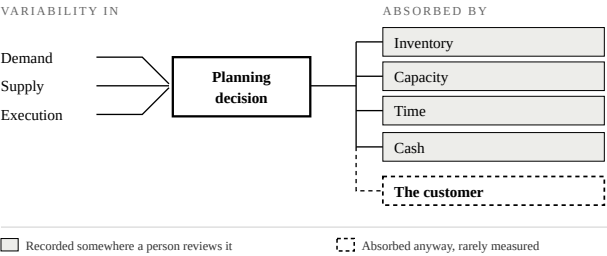


Figure 1. The five absorbers. Four of them sit in a system somebody reviews. The fifth usually sits in a customer’s patience, where the solver will find it and spend it.

FIELD LESSON

A company that cannot price a missed customer date is running its largest buffer off the books. Hand that company an optimizer and it will find the account and draw on it.

2. The number inside the buffer

The four weeks

A durables importer asked for a better demand forecast. Safety stock stood at four weeks of cover across effectively the whole catalog. Asked where four weeks came from, the planning manager said it was what they went to after the port thing. Three of the four people who had been in that room had left the company.

Safety stock under demand and lead time variability is standard:

$$SS = z \sqrt{\bar{L} \sigma_D^2 + \bar{D}^2 \sigma_L^2}$$

The second term is the reason to write it out. On their numbers, lead time variance carried most of the buffer, which put a demand forecasting project to work on the smaller term.

Their ERP recorded one lead time, purchase order to receipt, so lead time variance arrived as a single opaque number covering a chain of unrelated things. Broken into supplier confirmation, production, ocean transit, customs, and drayage into the DC, ocean transit came out among the better-behaved stages. Two carried long right tails: customs clearance on a subset of tariff codes, and appointment scheduling at their own building. They were financing inventory to cover a calendar problem forty feet from the receiving door.

LESSON. Work out which variance term the buffer is paying for before improving a forecast. Demand error shows up on a dashboard, so it gets requested first. The expensive term is often somewhere with no dashboard at all.

The ratio behind the service level

A specialty retailer held a 98% fill rate across the assortment. The number appeared in the supply chain scorecard, in vendor agreements, and in a board deck.

For a single-period stocking decision the optimal service level is the critical ratio:

$$q^* = C_u / (C_u + C_o)$$

with C_u the cost of one unit short and C_o the cost of one unit long. A 98% target is a standing claim that C_u divided by C_o equals 49. One unit short costs forty-nine times what one unit long costs.

We took that ratio to merchandising and finance for a dozen representative items. On the hero item that anchored a category and pulled attachments, nobody argued, and the honest number was arguably higher. On a commodity item with a near-identical substitute two feet away on the same shelf, the ratio was closer to three, since a stockout mostly moved the sale rather than losing it. On a seasonal item eight weeks from markdown, C_o climbed every week while the target held still.

The 98% had come from a director's previous employer, in a different category, roughly a decade earlier.

It was defensible for perhaps a tenth of the catalog. The rest of the assortment was being run on a number borrowed from items it had nothing in common with, and the cross-subsidy ran one direction:

substitutable goods held working capital that the margin drivers needed. They went to four cost tiers, a ratio for each, and a name against every ratio.

3. What the objective leaves out

Where nervousness comes from

A tier-one automotive supplier ran weekly MRP regeneration and got materially different plans out of nearly identical inputs. Released quantities moved, sequencing moved, and supplier releases followed. The planners had built the usual defenses: a set of maintained spreadsheets, a private set of rules, and the habit of running the official plan through personal judgment before anything left the building. One of them called his workbook the real plan, and he meant it literally.

The scoped project was forecast improvement, on the theory that a steadier forecast would produce a steadier plan.

The planning run re-optimized from scratch every cycle with no term for the cost of changing a commitment it had already made. Changing a plan costs setup and changeover, credibility with suppliers who staffed to the last release, labor scheduling, and the standing belief downstream that a release means something. The objective priced change at zero, so the solver spent it chasing small theoretical gains off noise in the inputs.

The fix is a term:

$$\min \sum_t c_t x_t + \lambda \sum_t |x_t - \tilde{x}_t|$$

where $\tilde{x}$ is the plan already communicated and λ prices the change. λ has a dollar value. The planners produced a defensible one in about an hour, because they had been paying it for years.

Churn dropped against a small and now-visible loss in theoretical optimality. That trade had been running the whole time, in the wrong direction, with nobody's signature on it.

LESSON. A solver spends whatever you leave out of the objective. Stability is the line item most often left out, and nervousness is the complaint that arrives in its place.

What happens when recourse is punished

An industrial distributor tracked expedited freight as an unfavorable variance and reviewed branch managers on it every month.

The branches responded rationally. Expedites got split below reporting thresholds, booked into other freight categories, arranged straight with carriers outside the TMS, or pre-empted by holding extra stock that no model knew about and nobody had authorized. The variance line improved every quarter while the freight bill held steady.

Expediting is recourse, the second-stage action in a problem where you commit, watch, then correct. Under real uncertainty some recourse is optimal. A distributor with near-zero expedite spend is over-buffered somewhere else and paying more for the privilege, in inventory that will look fine right up until the day it gets written down.

Moving expedite from a variance line to a budgeted recourse line with a unit price did two things. The spend became legitimate, so it got recorded. Recorded at lane and item level, it turned into the most useful dataset in the company: a map of where the buffers were wrong, updated daily by the people closest to the problem, at no cost. Shame an action and you lose the record of it while continuing to pay for it.

4. What the floor is already telling you

A metric with nowhere to go

A telecom field services group forecast spare parts and was managed on MAPE. Much of the catalog was intermittent: parts moving a handful of times a year, at a particular site, with long runs of zeros in between.

MAPE on intermittent series is undefined at zero and unstable near it, and the team knew this perfectly well. They reported at region and month, where the series aggregate into a stable denominator and the metric looks respectable. The stocking decision was made at site and week.

Nobody there would have called it gaming. They were responding correctly to a measurement with no honest room to improve at the level where the decision lived. The question in front of them was binary and local: stock this part at this site, or leave it. It turned on the cost of a truck roll against a part sitting on a shelf for two years, weighted by what the asset does to a customer while it is down.

LESSON. A metric with no honest room to improve gets moved by changing the level of aggregation. When a forecasting team spends more time defending a number than using it, check whether the number can move at the granularity of the decision.

The repair log

A consumer goods manufacturer had a working allocation optimizer and an override rate high enough that leadership had scoped the next phase as adoption: training, a redesigned interface, a communications plan. The override was applied in a downstream screen and kept nowhere. Once the plan published, the recommendation it replaced was gone.

Capturing overrides with a short reason code took a few days of work. Within a month the codes had collapsed into five buckets, three of which accounted for most of the volume: named-account allocation commitments made by sales, promotional protection for accounts in a live program, and a minimum-shipment convention with two customers whose receiving docks could not handle partial pallets. All of it real, all of it contractual, none of it in the model.

The planners were repairing the optimizer by hand, every day, for free, and the repair log was being deleted nightly.

LESSON. Overrides are labeled data about a missing constraint. Most adoption problems turn out to be specification problems, and the ones that really are about people get easier to see once the specification is right.

THE SIX ENGAGEMENTS

Setting	What they asked for	What was binding	What changed
Durables importer	A better demand forecast	Lead time variance, sitting in customs clearance and in appointment scheduling at their own DC	Lead time split into five stages; buffer re-derived against the stage carrying the tail
Specialty retailer	A 98% fill rate, more cheaply	One service target standing in for several hundred distinct cost ratios	Four cost tiers, a critical ratio for each, an owner against each ratio
Automotive tier one	A steadier forecast, to steady the plan	Zero price on changing a release the business had already communicated	Change penalty in the planning objective, with λ estimated by the planners
Industrial distributor	Less expedite spend	Recourse booked as failure, so it moved off the books instead of down	Expedite budgeted and priced; spend logged by lane and item
Telecom field services	Better spare parts forecast accuracy	A metric with no honest room to improve at the granularity of the stocking decision	Stocking framed as truck roll against holding cost, weighted by asset criticality
Consumer goods maker	Better planner adoption	Contractual and promotional commitments that were real, binding, and missing from the model	Overrides captured with reason codes, then promoted into constraints

5. Where we start

A first pass fits in about two weeks and touches nothing in production. It comes down to being stubborn about six things.

1. **One decision, named properly.** An order quantity, for an item, at a location, on a cadence. “Inventory management” is a department.
2. **One unit, traced end to end,** from the signal that triggered the decision through to the outcome and whoever saw it.

3. **Every buffer that touches it.** Who set it, what week, and what was going wrong at the time.

4. **The price behind each buffer, read back to its owner.** Somebody disagreeing with a number they own is the finding, and it happens more often than you would expect.
5. Where the variance goes when the buffer runs out, and whether anyone measures the landing.
6. **The spreadsheet, the override, and the side process.** They mark the edges of the model better than any interview.

A USEFUL TRIGGER

If your planners maintain a spreadsheet, the spreadsheet is the specification. Read it before you scope anything.

6. A different order of operations

Most supply chain analytics programs run in technology order: data, forecast, optimization, dashboard, adoption. Every step is defensible, which is part of why the sequence survives contact with a steering committee.

The order we work in now:

1. **Name the decision.** Who decides what, at what granularity, on what cadence.

2. **Locate the absorption.** Where variability lands today, and whose budget carries it.
3. **Price the consequences.** Shortage, excess, delay, change, expediting, and the customer’s patience.
4. Pick the simplest policy that survives repeated application.
5. **Design the recourse.** What happens when the plan is wrong, who may act, and at what price.
6. Close the loop, so outcomes and overrides reach the next cycle.
- Forecast improvement stays on the list and enters once somebody can name the decision it would change. On two of these six engagements it produced a real forecasting project, better specified for having been asked second.

THE CORE TAKEAWAY

Somebody set that number. Find out who, find out what week it was, and ask whether the conditions that produced it are still in the building.

WHAT A SUPPLY CHAIN UNCERTAINTY AUDIT TESTS

Decision	Is the recurring choice defined at the granularity, timing, and unit at which somebody actually makes it?
Absorption	Where does variability land today, across inventory, capacity, time, cash, and the customer, and whose budget carries it?
Prices	Are shortage, excess, delay, change, and expediting priced, or standing behind a target nobody has re-derived?
Rules	Which constraints are physical or contractual, and which are convention that has never been tested?
Recourse	When the plan is wrong, who may act, at what price, and does the action survive in a form the next cycle can use?
Learning	Do realized outcomes and overrides reach the next decision, or stop at a report?

ABOUT BIT BROS

Bit Bros is a boutique applied economics firm working on supply chain and decision systems, at the intersection of optimization, simulation, forecasting, and AI. Founder experience includes Amazon, Toyota, Gurobi, and AT&T. We work on the operating mechanism around the model: how decisions get specified, priced, tested, deployed, and revisited.

The Decision Factory (DeJans and Elam, 2026) tells this story as a novel. Field Note 01, *When Better Models Stop Creating Better Decisions*, gives the general decision-system version. This paper is the supply chain case.

SUPPLY CHAIN UNCERTAINTY AUDIT

For teams about to fund a forecasting, inventory optimization, or planning system initiative, we run the diagnostic above before significant engineering is committed. It is short, it is narrow on purpose, and it ends in a prioritized list of what to fix, simplify, test, or leave alone.

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